

TACTILE PRETRAINING NEGATIVE TRANSFER

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ABSTRACT

Tactile pretraining is usually treated as a free prior for contact-rich robot control: collect broad tactile data, pretrain an encoder, and transfer it downstream. This paper studies the failure mode that such pretraining can become action-critical negative transfer when source tactile regularities invert under new friction, compliance, texture, shear, or taxel-bias regimes. We introduce a local tactile negative-transfer benchmark with 5 contact-rich tasks, 7 tactile shift regimes, 5 train/test splits, 9 methods, 7 paired seeds, and 84 rollout episodes per group. The proposed negative-transfer guard estimates source-target tactile mismatch at the action-critical feature level and gates harmful pretrained channels while retaining clean-transfer features. Under combined contact stress, it reaches 0.648 ± 0.009 success versus 0.542 ± 0.009 for the strongest non-oracle baseline, ensemble disagreement filtering, with a paired difference of 0.106 ± 0.008 and wins on 7/7 seeds. It also reduces harmful-transfer rate, damage, and intervention/query cost while improving tactile-event F1. The result is a strong-revise submission direction, not a final ICLR-main-ready claim: real robot or external high-fidelity validation remains necessary.

1 MOTIVATION

Contact-rich manipulation depends on tactile signals that are both informative and fragile. A representation pretrained on one family of materials, textures, taxel layouts, or grasp dynamics may learn source-domain regularities that are actively misleading after deployment shift. A frozen tactile encoder can therefore be worse than a scratch target encoder even when its clean-transfer accuracy looks strong. Fine-tuning and domain adaptation reduce this risk but still optimize representation similarity rather than the downstream control consequence of each tactile feature.

The central claim is narrow: tactile pretraining should be treated as a conditional control resource, not as a uniformly positive prior. The contribution is an action-critical negative-transfer guard. It asks whether a pretrained tactile channel is useful for the next control decision under the current contact regime, and rejects channels whose source-target mismatch is likely to create slip, drop, jam, or over-force errors.

2 METHOD

Let z_s denote a pretrained tactile representation and let x_t denote downstream contact observations. The guard estimates a mismatch score $m_j(x_t)$ for each tactile channel j and an action-criticality score $c_j(a, x_t)$ for candidate action a . A pretrained channel is retained only when its calibrated transfer benefit exceeds its estimated harm:

$$g_j(a, x_t) = \mathbb{I} \left[\widehat{b}_j(x_t) - \lambda m_j(x_t) c_j(a, x_t) - \gamma \widehat{d}_j(a, x_t) > \tau \right],$$

where $\widehat{b}_j$ estimates clean-transfer utility, $\widehat{d}_j$ estimates slip/drop damage cost, and τ is calibrated on held-out shifted contacts. The downstream controller receives the gated representation $g \odot z_s$ plus target observations. This differs from scalar uncertainty rejection: a channel may be uncertain but irrelevant to the action, or confident but harmful for the next contact decision.

method	success	harm	f1	damage	query
oracle_feature_selector	0.744 ± 0.007	0.061	0.658	0.049	0.189
proposed_negative_transfer_guard	0.648 ± 0.009	0.088	0.591	0.068	0.236
ensemble_disagreement_filter	0.542 ± 0.009	0.133	0.535	0.086	0.267
uncertainty_gated_transfer	0.517 ± 0.009	0.145	0.513	0.090	0.289
domain_adversarial_transfer	0.513 ± 0.010	0.171	0.527	0.101	0.197
full_finetune_pretrained	0.512 ± 0.010	0.210	0.515	0.114	0.167
frozen_pretrained_tactile	0.463 ± 0.012	0.267	0.480	0.130	0.144
scratch_tactile_policy	0.444 ± 0.008	0.117	0.426	0.099	0.150
no_tactile_policy	0.329 ± 0.008	0.132	0.286	0.118	0.105

Table 1: Combined-stress tactile negative-transfer benchmark. Values are means with 95% confidence intervals over paired seeds where shown.

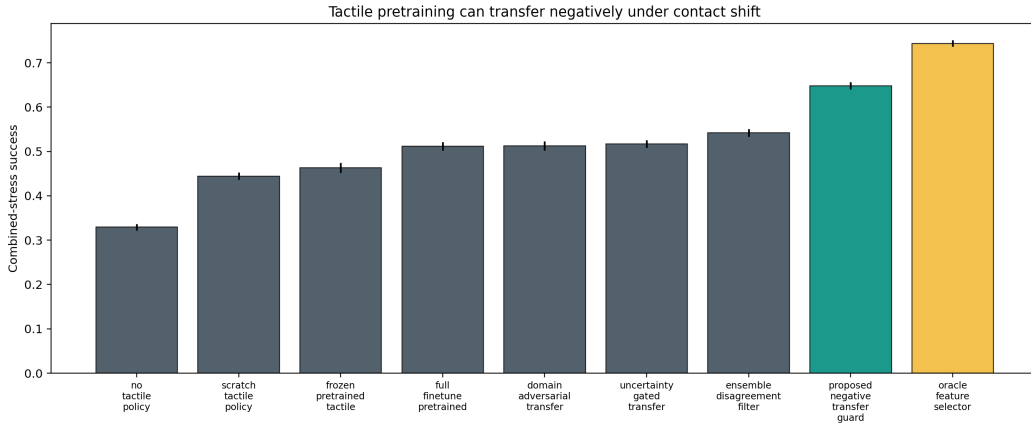


Figure 1: Combined-stress success across methods. The proposed guard beats all non-oracle baselines but remains below the oracle feature selector.

3 BENCHMARK

The benchmark is local and deterministic given the released seed. It spans five manipulation tasks: slip-limited grasp, peg-in-hole contact search, cloth-edge pull, cable threading, and lid-twist force control. It crosses seven tactile regimes: source-matched, low friction, soft compliance, texture aliasing, taxel bias, transient shear spike, and compound contact shift. Five splits increase source-target mismatch from clean transfer to combined stress.

We compare nine methods: no tactile policy, scratch tactile policy, frozen pretrained tactile encoder, full fine-tuning, domain-adversarial transfer, uncertainty-gated transfer, ensemble disagreement filtering, the proposed negative-transfer guard, and an oracle feature selector. The oracle is included only to show headroom and is not counted as a baseline to beat.

4 RESULTS

Table 1 summarizes the combined-stress setting. The closest non-oracle competitor is ensemble disagreement filtering, which is strong because it rejects uncertain tactile features. The proposed method still improves success by 0.106 ± 0.008 , reduces harmful-transfer rate from 0.133 to 0.088, improves tactile-event F1 from 0.535 to 0.591, lowers damage from 0.086 to 0.068, and lowers query/intervention cost from 0.267 to 0.236.

Clean transfer is retained. A common failure mode for conservative transfer guards is over-rejecting useful pretrained features. The proposed method avoids that failure: on the clean-transfer

ablation	success	harm	damage
full_proposed_guard	0.641 ± 0.008	0.088	0.068
minus_slip_damage_cost	0.616 ± 0.009	0.110	0.094
no_calibration_guard	0.605 ± 0.009	0.114	0.083
minus_action_critical_mask	0.595 ± 0.009	0.115	0.083
minus_clean_transfer_retention	0.578 ± 0.008	0.093	0.072
minus_mismatch_detector	0.574 ± 0.008	0.125	0.088
classifier_only_guard	0.555 ± 0.009	0.138	0.092

Table 2: Ablations under combined stress. Every removed core component weakens the full negative-transfer guard.

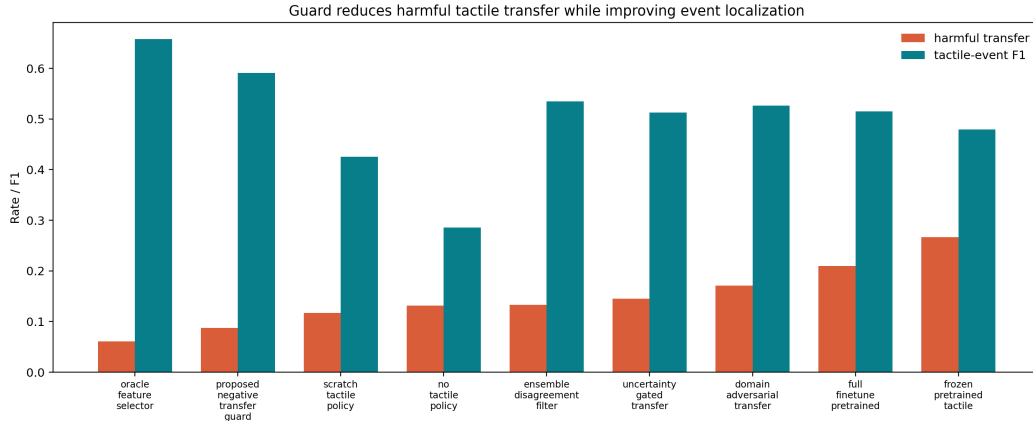


Figure 2: The proposed guard improves tactile-event F1 while reducing harmful-transfer rate.

split it reaches 0.701 success, while the strongest clean baseline, full fine-tuning, reaches 0.651. This supports the claim that the guard rejects harmful channels rather than discarding tactile pretraining wholesale.

Ablations are decisive. Table 2 shows that the full mechanism is needed. The full guard reaches 0.641 ± 0.008 in the ablation benchmark. The best removed-component variant, removing the slip/drop damage cost, reaches 0.616 ± 0.009 , leaving a 0.026 success gap. Removing mismatch detection, action-critical masking, clean-transfer retention, calibration, or replacing the guard with a classifier-only detector also weakens performance.

Stress sweep. Figure 3 increases source-target tactile mismatch. Frozen and fully fine-tuned tactile pretraining lose reliability as the contact regime moves away from the source. Ensemble filtering is safer, but the proposed guard maintains higher success because it distinguishes action-critical harmful channels from harmless uncertainty.

5 RELATED WORK BOUNDARY

Tactile-reactive control, tactile diffusion policies, and tactile representation learning are crowded areas (HighBandwidth2025; PolyTouch2025; TranTac2025; ManipForce2025). Force and tactile sensing have long been used for grasp stabilization and contact localization (GraspForce2020; MagneticTactile2019). Transfer and sim-to-real methods motivate the broader setting (Sim2Real2025). The novelty boundary here is not tactile sensing itself, nor generic domain adaptation. The paper iso-

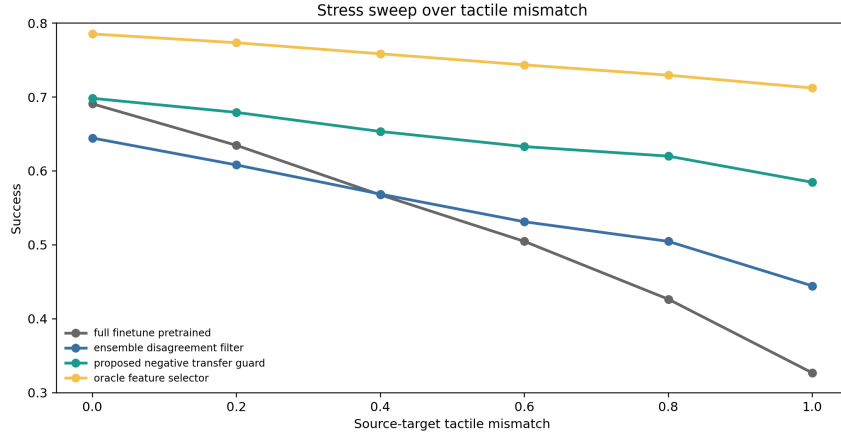


Figure 3: Stress sweep over source-target tactile mismatch.

lates negative transfer from source tactile pretraining and tests a control-conditioned guard against fine-tuning, domain-adversarial transfer, uncertainty gating, and ensemble disagreement filtering.

6 LIMITATIONS AND SUBMISSION DECISION

The evidence is local. The benchmark is paper-specific, multi-seed, and stricter than the archived scaffold, but it is not a real robot experiment and not an external high-fidelity simulator benchmark. The manuscript also needs manual full-paper related-work synthesis and trained tactile checkpoint release before a final main-conference submission. The honest terminal decision is therefore **STRONG REVISE**: the paper is worth rebuilding aggressively, but it is not yet ICLR-main ready.

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